Lab11

Siddharth Bailkeri

Figure 18:

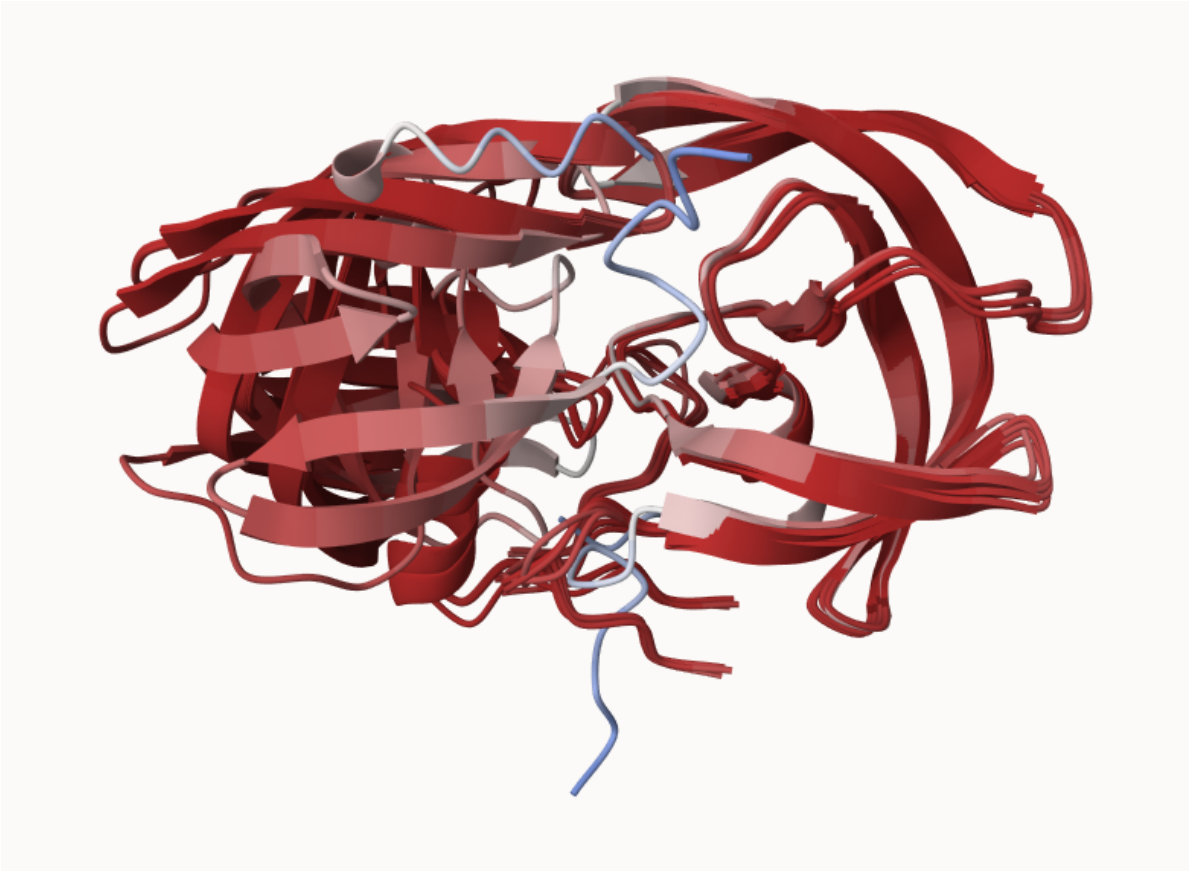


Figure 1: Superposed Mol* rendering of 5 sequences of HIV-1 protease, coloured by pLDDT scores. Sequences were obtained through alphafold/colabfold

8. Custom analysis of resulting models

```
results_dir <- "hivprhomodimer_23119"
pdb_files <- list.files(path=results_dir,
                        pattern="*.pdb",
                        full.names = TRUE)

# Print our PDB file names
basename(pdb_files)
```

```
[1] "hivprhomodimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_seed_000.pdb"
[2] "hivprhomodimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_4_seed_000.pdb"
[3] "hivprhomodimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_1_seed_000.pdb"
[4] "hivprhomodimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_5_seed_000.pdb"
[5] "hivprhomodimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb"
```

```
library(bio3d)

pdbs <- pdbaln(pdb_files, fit = TRUE, exefile = "msa")
```

```
Reading PDB files:
hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_2_s
hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_4_s
hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_1_s
hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_5_s
hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_s
.....

Extracting sequences

pdb/seq: 1   name: hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_001_alphafold2_mu
pdb/seq: 2   name: hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_002_alphafold2_mu
pdb/seq: 3   name: hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_003_alphafold2_mu
pdb/seq: 4   name: hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_004_alphafold2_mu
pdb/seq: 5   name: hivprhomodimer_23119/hivprhomodimer_23119_unrelaxed_rank_005_alphafold2_mu
```

```
pdbs
```

```
                                    1        .        .        .        .        50
```

```
[Truncated_Name:1]hivprhomod   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:2]hivprhomod   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:3]hivprhomod   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:4]hivprhomod   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
[Truncated_Name:5]hivprhomod   PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGI
                               **************************************************
                               1        .         .         .         .         50

                              51        .         .         .         .        100
[Truncated_Name:1]hivprhomod   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:2]hivprhomod   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:3]hivprhomod   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:4]hivprhomod   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
[Truncated_Name:5]hivprhomod   GGFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFP
                               **************************************************
                              51        .         .         .         .        100

                             101        .         .         .         .        150
[Truncated_Name:1]hivprhomod   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:2]hivprhomod   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:3]hivprhomod   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:4]hivprhomod   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
[Truncated_Name:5]hivprhomod   QITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIG
                               **************************************************
                             101        .         .         .         .        150

                             151        .         .         .         .       198
[Truncated_Name:1]hivprhomod   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:2]hivprhomod   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:3]hivprhomod   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:4]hivprhomod   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
[Truncated_Name:5]hivprhomod   GFIKVRQYDQILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNF
                               **************************************************
                             151        .         .         .         .       198

Call:
  pdbaln(files = pdb_files, fit = TRUE, exefile = "msa")

Class:
  pdbs, fasta

Alignment dimensions:
  5 sequence rows; 198 position columns (198 non-gap, 0 gap)
```

```
+ attr: xyz, resno, b, chain, id, ali, resid, sse, call
```

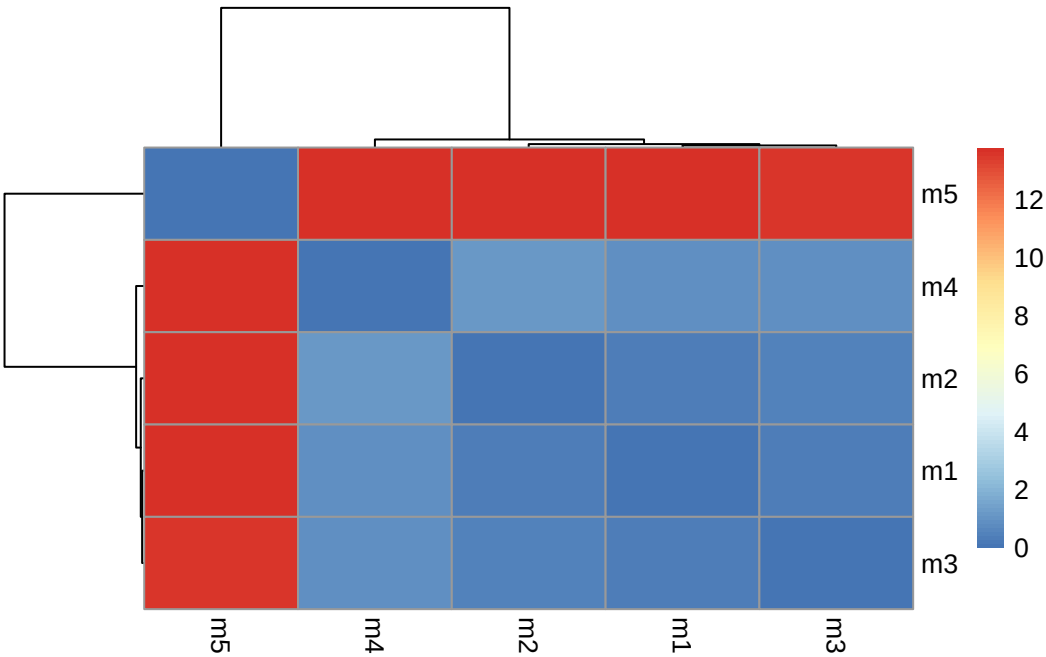
```r
rd <- rmsd(pdbs, fit = T)
```

```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 13.795
```

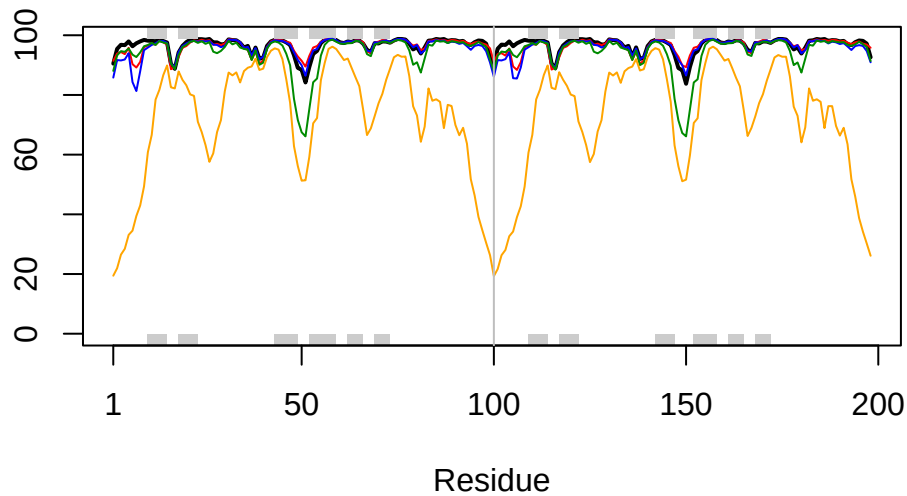
```r
library(pheatmap)
colnames(rd) <- paste0("m", 1:5)
rownames(rd) <- paste0("m", 1:5)
pheatmap(rd)
```




```r
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```r
plotb3(pdbs$b[1, ], type = "l", lwd = 2, sse= pdb)
points(pdbs$b[2, ], type = "l", col = "red")
points(pdbs$b[3, ], type = "l", col = "blue")
points(pdbs$b[4, ], type = "l", col = "green4")
points(pdbs$b[5, ], type = "l", col = "orange")
abline(v = 100, col = "grey")
```



Finding the core:

```r
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 100.201
core size 196 of 198  vol = 94.434
core size 195 of 198  vol = 89.229
core size 194 of 198  vol = 83.206
core size 193 of 198  vol = 80.468
core size 192 of 198  vol = 78.538
core size 191 of 198  vol = 76.265
core size 190 of 198  vol = 74.273
core size 189 of 198  vol = 72.113
core size 188 of 198  vol = 70.014
core size 187 of 198  vol = 68.161
```

```
core size 186 of 198  vol = 66.667
core size 185 of 198  vol = 65.366
core size 184 of 198  vol = 64.163
core size 183 of 198  vol = 63.182
core size 182 of 198  vol = 63.107
core size 181 of 198  vol = 63.131
core size 180 of 198  vol = 63.993
core size 179 of 198  vol = 64.314
core size 178 of 198  vol = 64.41
core size 177 of 198  vol = 64.919
core size 176 of 198  vol = 65.089
core size 175 of 198  vol = 65.419
core size 174 of 198  vol = 65.019
core size 173 of 198  vol = 65.321
core size 172 of 198  vol = 65.044
core size 171 of 198  vol = 64.057
core size 170 of 198  vol = 63.193
core size 169 of 198  vol = 61.892
core size 168 of 198  vol = 60.493
core size 167 of 198  vol = 59.191
core size 166 of 198  vol = 57.83
core size 165 of 198  vol = 56.382
core size 164 of 198  vol = 55.044
core size 163 of 198  vol = 53.655
core size 162 of 198  vol = 52.229
core size 161 of 198  vol = 51.146
core size 160 of 198  vol = 49.964
core size 159 of 198  vol = 48.815
core size 158 of 198  vol = 47.493
core size 157 of 198  vol = 45.862
core size 156 of 198  vol = 44.987
core size 155 of 198  vol = 43.806
core size 154 of 198  vol = 42.71
core size 153 of 198  vol = 41.776
core size 152 of 198  vol = 41.067
core size 151 of 198  vol = 40.021
core size 150 of 198  vol = 39.051
core size 149 of 198  vol = 38.014
core size 148 of 198  vol = 36.152
core size 147 of 198  vol = 35.265
core size 146 of 198  vol = 34.26
core size 145 of 198  vol = 33.302
core size 144 of 198  vol = 32.803
```

```
core size 143 of 198  vol = 32.615
core size 142 of 198  vol = 30.867
core size 141 of 198  vol = 30.109
core size 140 of 198  vol = 29.316
core size 139 of 198  vol = 28.15
core size 138 of 198  vol = 27.065
core size 137 of 198  vol = 25.639
core size 136 of 198  vol = 24.471
core size 135 of 198  vol = 23.316
core size 134 of 198  vol = 21.712
core size 133 of 198  vol = 20.363
core size 132 of 198  vol = 19.768
core size 131 of 198  vol = 18.755
core size 130 of 198  vol = 17.882
core size 129 of 198  vol = 17.185
core size 128 of 198  vol = 16.511
core size 127 of 198  vol = 15.977
core size 126 of 198  vol = 15.208
core size 125 of 198  vol = 14.437
core size 124 of 198  vol = 13.509
core size 123 of 198  vol = 12.856
core size 122 of 198  vol = 12.43
core size 121 of 198  vol = 12.007
core size 120 of 198  vol = 11.557
core size 119 of 198  vol = 10.8
core size 118 of 198  vol = 10.15
core size 117 of 198  vol = 9.55
core size 116 of 198  vol = 9.014
core size 115 of 198  vol = 8.283
core size 114 of 198  vol = 7.519
core size 113 of 198  vol = 6.947
core size 112 of 198  vol = 6.37
core size 111 of 198  vol = 6.025
core size 110 of 198  vol = 5.681
core size 109 of 198  vol = 5.363
core size 108 of 198  vol = 5.022
core size 107 of 198  vol = 4.771
core size 106 of 198  vol = 4.405
core size 105 of 198  vol = 4.109
core size 104 of 198  vol = 3.976
core size 103 of 198  vol = 3.75
core size 102 of 198  vol = 3.584
core size 101 of 198  vol = 3.429
```

```
 core size 100 of 198  vol = 3.139
 core size 99 of 198  vol = 2.943
 core size 98 of 198  vol = 2.755
 core size 97 of 198  vol = 2.365
 core size 96 of 198  vol = 1.968
 core size 95 of 198  vol = 1.799
 core size 94 of 198  vol = 1.683
 core size 93 of 198  vol = 1.295
 core size 92 of 198  vol = 0.985
 core size 91 of 198  vol = 0.893
 core size 90 of 198  vol = 0.697
 core size 89 of 198  vol = 0.606
 core size 88 of 198  vol = 0.445
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol= 0.5)
```

```
# 89 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     8  48     41
2    52  78     27
3    80  97     18
4    99  99      1
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath = "corefit_structures")
```

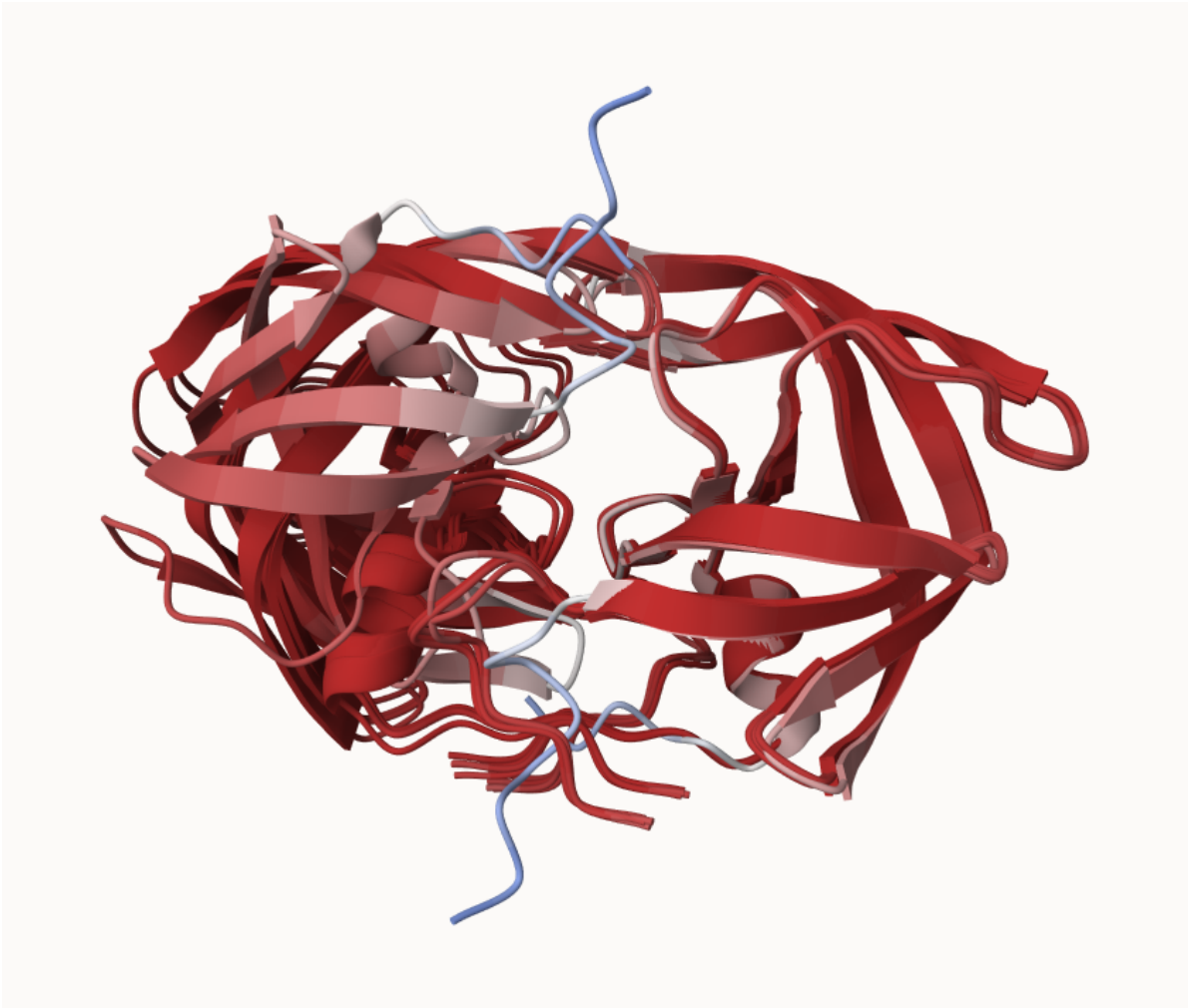
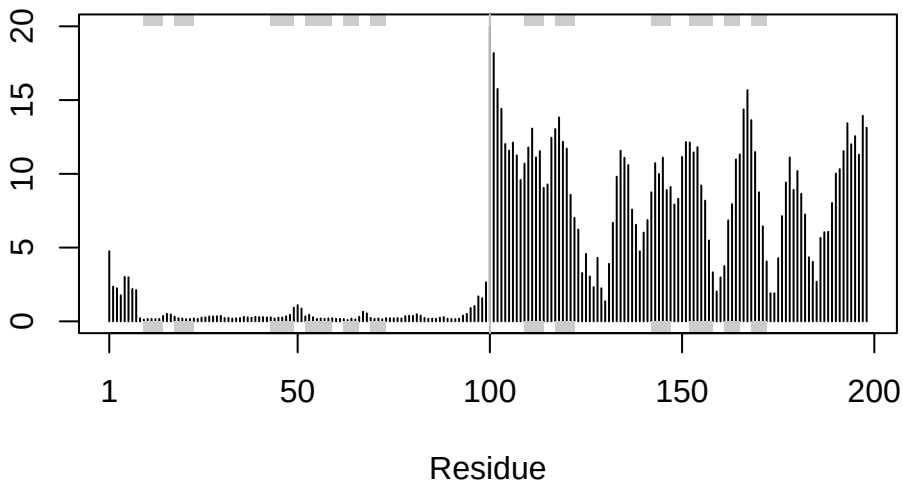


Figure 2: Core superposed structure colored by pLDDT score

Examining RMSF

```
rf <- rmsf(xyz)
plotb3(rf, sse = pdb)
abline(v = 100, col = "gray", ylab = "RMSF")
```



Predicted Alignment Error for domains

```
library(jsonlite)
pae_files <- list.files(path = results_dir,
                        pattern = ".*model.*\\.json",
                        full.names = TRUE)
```

```
pae1 <- read_json(pae_files[1], simplifyVector = TRUE)
pae5 <- read_json(pae_files[5], simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```
head(pae1$plddt)
```

```
[1] 90.50 95.44 96.81 96.69 97.94 96.31
```

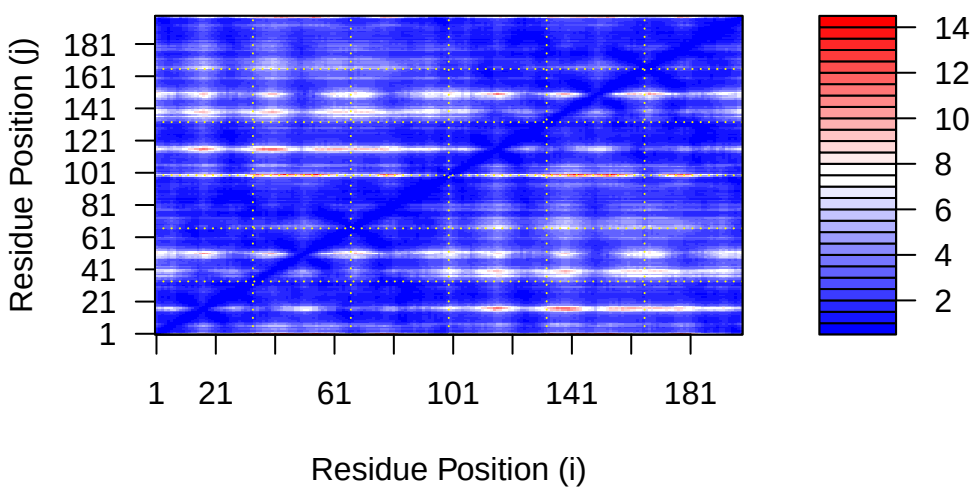
```r
pae1$max_pae
```

```
[1] 14.29688
```

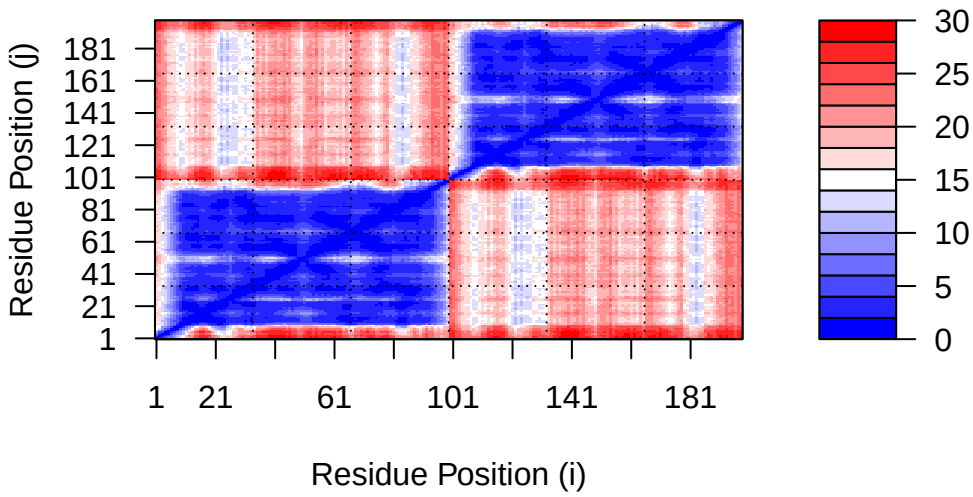
```r
pae5$max_pae
```

```
[1] 29.8125
```

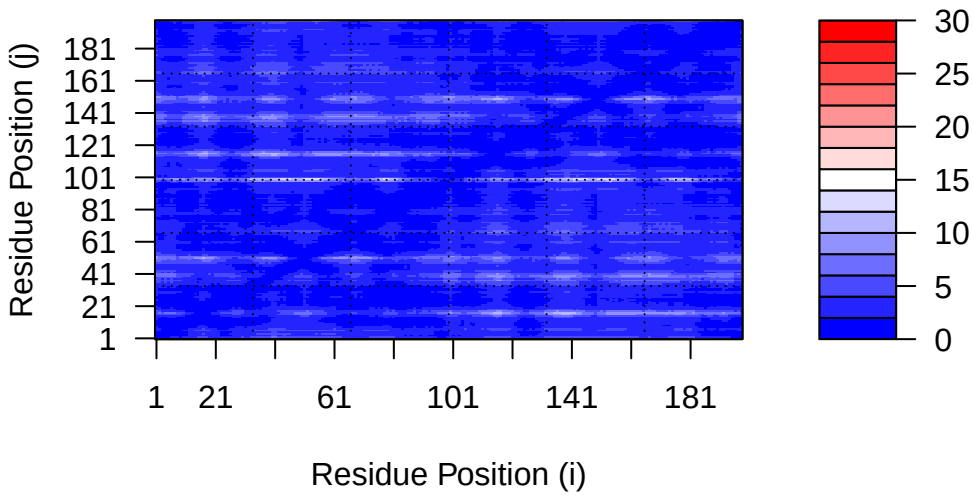
```r
plot.dmat(pae1$pae,
          xlab = "Residue Position (i)",
          ylab = "Residue Position (j)")
```




```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```




```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue conservation from alignment file

```
aln_file <- list.files(path=results_dir,
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivprhomodimer_23119/hivprhomodimer_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

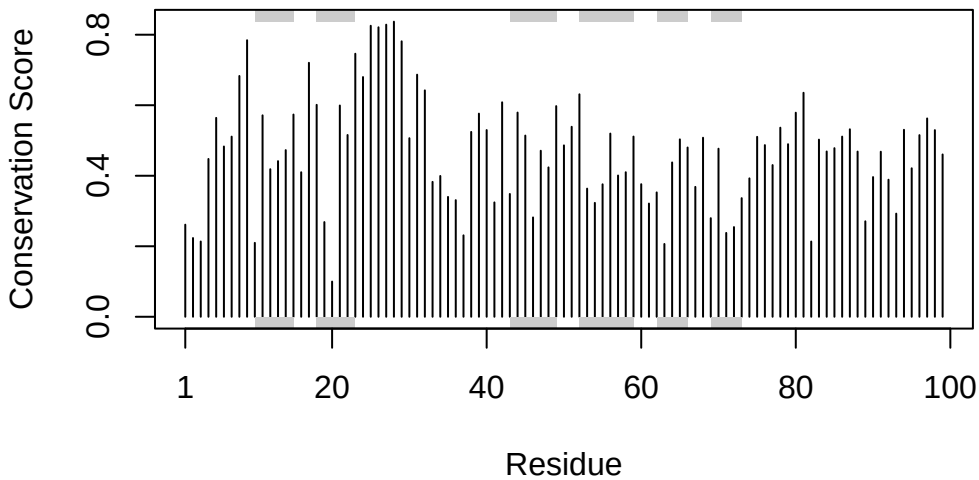
```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

```
dim(aln$ali)
```

```
[1] 5397  132
```

```r
sim <- conserv(aln)
```

```r
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```




```r
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

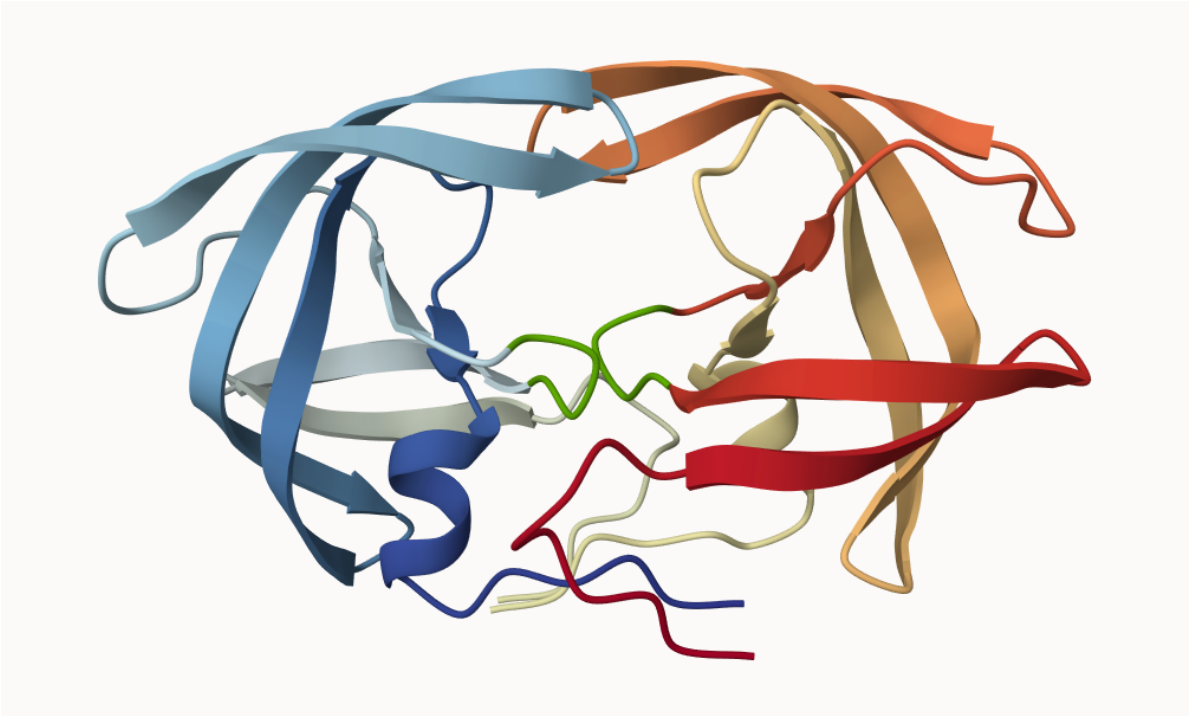


Figure 3: Mol* rendering of top-ranked dimer colored by element index. Conserved DGTA active site motif on both chains shown colored in green.